

A scattering matrix without a space

The shifted Weil transfer as an invariant of the rank, the interval on which its criterion has content, and why the lattice's failure leaves no trace in it

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Abstract

The shifted Weil family turns the Riemann hypothesis into a statement about causal all-pass filters: for each shift $\omega > 0$ the transfer $K_\omega(p) = \xi(\frac{1}{2} + p - \omega)/\xi(\frac{1}{2} + p + \omega)$ is unimodular on the critical line, and its compressions to windows of every length are contractions if and only if no zero lies further than ω from that line. Its Markov part $\tilde{K}_\omega = \Lambda(\frac{1}{2} + p - \omega)/\Lambda(\frac{1}{2} + p + \omega)$ is completely monotone, and every one of its factors is d -dimensional as a formula for $d = 2\omega$. This manuscript asks what that d is. We show first that it is a *rank*: for a covolume-one lattice $L \subset \mathbb{R}^m$ the primitive Epstein zeta $E_L^*(s) = \sum_{v \in \text{prim}} |v|^{-2s}$ satisfies $\Lambda(2s)E_L^*(s) = \Lambda(m - 2s)E_{L^*}^*(m/2 - s)$, so its reflection ratio is independent of L and is an invariant of the rank alone — and with $m = d + 1$ that ratio is exactly $\tilde{K}_\omega$, whose own reflection $p \mapsto -p$ is the Epstein functional equation. Every integer $d \geq 1$ is realized, as rank $d + 1$; the family's range $\omega \leq \frac{1}{2}$ is the first gap $m \in (1, 2]$ of that sequence. Everything then turns on the sign of $a = \frac{1}{2} - \omega = \frac{2-m}{2}$. We show that at $a = 0$ three things happen at once: the second pole of $\tilde{K}_\omega$ — which is the *archimedean* factor's, while the first is the comb's — crosses into the right half-plane; the rational correction $R_\omega = B_b \cdot \frac{p+a}{p-a}$ turns from a ratio of Blaschke factors with a right half-plane pole into the inner product $B_b^* B_c$, $c = \omega - \frac{1}{2}$; and the poles of K_ω leave the closed right half-plane, which makes every compression a contraction *unconditionally, on the strength of the Euler product alone*. Complete monotonicity survives the crossing, by two groupings of the Gamma factors that are exchanged there, so the decomposition continues past the wall with its positivity intact — but the criterion does not: it is vacuous for $\omega \geq \frac{1}{2}$ and has arithmetic content on exactly $\omega \in (0, \frac{1}{2})$, that is $m \in (1, 2)$, where it is equivalent to a zero-free strip of half-width ω . The Euler product is therefore the criterion's *value* at rank two and the Riemann hypothesis is its *derivative* at rank one, where the transfer degenerates to the identity and the first-order defect is the localized Weil form. Inside the gap no lattice exists: the point count's unique interpolation is the q -expansion of θ^m , and on the gap $(k, k + 1)$ the binomial $\binom{m}{j}$ is positive through $j = k + 1$ and first negative at $j = k + 2$, so the first coefficient that is *forced* negative sits at the least integer needing $k + 2$ positive squares — 2, 3, 7, and none at all once $k \geq 3$, because Lagrange's four-square theorem bounds that least number by four. The two failures share the interval $m \in (1, 2)$ and nothing else: the point count fails on $(2, 3)$ as well, where the criterion is already vacuous, and at $m = 3$ the point count is completely monotone while the transfer's rational factor is not. The agreement is the first gap, forced by both of the criterion's endpoints lying at the two smallest ranks. What separates them is where the rank sits: in the transfer it is always an *exponent* — $(2 \sinh \tau)^{\omega-1}$, $(1 - e^{-t})^{\beta-\alpha-1}$, p^{-d} — so each positivity is a local inequality, at one delay or at one prime, that continuation cannot break; in the point count it is a *binomial index*, where positivity is a global cancellation. The transfer's positivity was never inherited from a lattice, and the lattice's failure leaves no trace in it. At fractional d there is a scattering matrix without a space, and d is a parameter. Nothing here constructs a source, proves positivity of the Weil form, or assumes the Riemann hypothesis.

1 Introduction

1.1 The family, and the question this manuscript answers

Weil's criterion turns the Riemann hypothesis into a positivity statement: a certain quadratic form Q_L , built from the primes below e^L and an archimedean term, is positive semidefinite for every window length L if and only if every nontrivial zero of ζ lies on the critical line. The *shifted* family [3] replaces that single form by a one-parameter deformation. For $\omega > 0$ put $s = \frac{1}{2} + p$ and

$$K_\omega(p) = \frac{\xi(s - \omega)}{\xi(s + \omega)}.$$

The functional equation makes $|K_\omega| = 1$ on the imaginary axis, so K_ω is an all-pass filter; its causal realization V_ω compressed to a window of length L is a contraction for every L precisely when no zero lies further than ω from the critical line; and the defect of that compression is, to first order in ω , twice the Weil form. The Riemann hypothesis is therefore the assertion that the criterion holds for every $\omega > 0$, and the family interpolates between something and it.

The parent manuscript [2] factored the transfer. Writing $\xi(u) = \frac{1}{2}u(u-1)\Lambda(u)$, the ratio $\widetilde{K}_\omega = \Lambda(s - \omega)/\Lambda(s + \omega)$ is completely monotone — the Laplace transform of an explicit positive measure, a Beta-distributed continuous delay convolved with a comb at the integers — and what is left is a rational all-pass factor forced by unimodularity. It also found that every piece of $\widetilde{K}_\omega$ is d -dimensional as a formula, for $d = 2\omega$: the archimedean factor is a Riesz integral over $\mathbb{R}^d$, the comb's weights are the Jordan totient J_d , and the residue at the Blaschke pole is the volume term of a lattice of rank $d + 1$ [4, 5, 6]. The question that opened the present investigation is the obvious one.

Problem 1.1. Is $d = 2\omega$ a dimension, or a parameter?

This manuscript answers it: **a parameter**. The route runs through four statements, and the fourth is the one worth keeping.

1.2 What is proved

The transfer is an invariant of the rank (Section 3). For a covolume-one lattice $L \subset \mathbb{R}^m$ let $E_L^*(s) = \sum_v \text{prim} |v|^{-2s}$ be the primitive Epstein zeta. Poisson summation and the factorization $Z_L = \zeta(2s)E_L^*$ give $\Lambda(2s)E_L^*(s) = \Lambda(m - 2s)E_{L^*}^*(m/2 - s)$, so the reflection ratio is independent of L and depends on the rank alone — and with $m = d + 1$ and $2s = p + b$ that ratio is exactly $\widetilde{K}_\omega$. So the transfer is the scattering matrix of the unimodular lattices of rank $d + 1$, at *every* integer $d \geq 1$, and its own reflection $p \mapsto -p$ is the Epstein functional equation. The family's range $\omega \in (0, \frac{1}{2}]$ is the first gap $m \in (1, 2]$ of that sequence, with the modular surface at its right endpoint.

The wall is the sign of a (Section 4). With $a = \frac{1}{2} - \omega = \frac{2-m}{2}$ and $c = -a$, three things happen at $a = 0$, that is at $m = 2$. The second pole of $\widetilde{K}_\omega$ crosses into the right half-plane — and it is the *archimedean* factor's pole, the one from Γ , while the first is the comb's, the one from ζ (Table 1). The rational factor $R_\omega = B_b \cdot \frac{p+a}{p-a}$ turns from a ratio B_b/B_a with a right half-plane pole into the inner product $B_b B_c$ (Proposition 4.2). And complete monotonicity *survives*: $\widehat{K}_\omega = \frac{p+a}{p-a} \widetilde{K}_\omega$ is completely monotone at every real $\omega > 0$, by two groupings of the Gamma factors that are exchanged at the wall, the second being $\widehat{K}_\omega^\Gamma = \pi^\omega \frac{\Gamma(z+c)}{\Gamma(z+\omega)} \frac{\Gamma(z+1)}{\Gamma(z+c+1)}$ with $z = \frac{p+a}{2}$ (Proposition 4.3). The decomposition continues past the family's range with its positivity intact.

But the criterion does not (Section 5). The poles of K_ω sit at $p = \rho - b$, so they leave the closed right half-plane as soon as $a < 0$. For $\omega \geq \frac{1}{2}$, therefore, every compression $V_{\omega,L}$ is a contraction *unconditionally*, on the strength of the Euler product alone (Proposition 5.1): the criterion is vacuous there, and has arithmetic content on exactly $\omega \in (0, \frac{1}{2})$, where it is equivalent to a zero-free strip of half-width ω (Corollary 5.3). So the Euler product is the criterion's *value* at rank two, and the Riemann hypothesis is its *derivative* at rank one, where $E_{\mathbb{Z}}^* \equiv 2$, $\widetilde{K}_0 = 1$, $V_0 = I$, and the first-order defect is the Weil form (Table 3). The family runs from the easiest theorem about ζ to the hardest conjecture, and it runs downward in rank.

Inside the gap there is no lattice, and the reason is Lagrange (Section 6). The point count of $\mathbb{Z}^m$ has a unique interpolation, the q -coefficients of θ^m , polynomial in m . Writing $\theta^m = \sum_j \binom{m}{j} (\theta - 1)^j$ and letting $R_j(n)$ count ordered representations of n as j positive squares, one has $r_m(n) = \sum_j \binom{m}{j} 2^j R_j(n)$ with $R_j(n) = 0$ for $j > n$. On the gap $(k, k+1)$ the binomial is positive through $j = k+1$ and first negative at $j = k+2$ (Proposition 6.2), so $r_m(n) > 0$ for free when $n \leq k+1$ (Proposition 6.3) and the first coefficient that can be *forced* negative sits at N_k , the least integer needing $k+2$ positive squares. Those are $N_0 = 2$, $N_1 = 3$, $N_2 = 7$ — *and nothing at all for $k \geq 3$, because Lagrange's four-square theorem bounds that least number by four* (Propositions 6.4 and 6.5). In particular $r_m(3) = \frac{4}{3}m(m-1)(m-2) < 0$ throughout $m \in (1, 2)$: at fractional d there is no positive measure and hence no point set.

And the two failures are unrelated (Section 7). Both the criterion's content and the point count's failure live on $m \in (1, 2)$, which invites the guess that they are one fact. They are not. The point count fails on $(2, 3)$ as well, where the criterion is already vacuous; and at $m = 3$ the point count is completely monotone while the transfer's rational factor is not (Corollary 7.3). The agreement is the first gap and nothing more: the criterion's window has its endpoints at the two smallest ranks, and the first gap of the rank sequence is $(1, 2)$ by definition (Remark 7.4). What separates the two sides is where the rank sits. In the transfer it is always an *exponent* — $(2 \sinh \tau)^{\omega-1}$ in the archimedean kernel, $(1 - e^{-t})^{\beta-\alpha-1}$ in each Beta density, q^{-d} in each Euler factor of the comb — so each positivity is a *local* inequality, at one delay or at one prime, which continuation in the parameter cannot break. In the point count it is a *binomial index*, where positivity is a global cancellation (Proposition 7.5). The transfer's positivity was therefore never inherited from a lattice, and the lattice's failure leaves no trace in it; in particular there is no factorization of the transfer exhibiting which factor loses positivity at fractional d , because none does (Corollary 7.7).

1.3 What this is not

Nothing here constructs a source, proves any positivity of the Weil form, or assumes the Riemann hypothesis; no statement below is conditional on it. Nothing here is a new bound on the zeros: Proposition 5.1 is a statement that a criterion is empty, not that anything is true of ζ beyond the Euler product. The contraction numbers of Section 5.3 are a calibration of an inherited instrument at shifts where the answer is known in advance, and they are one-sided in the direction that does *not* certify contraction (Remark 5.5). The group-theoretic language of Corollary 3.3 is the standard reading of verified formulas and is labelled as such. The two statements about sums of squares in Remark 6.7 are conjectures and are used nowhere.

Section 2 fixes notation — including a warning, Remark 2.1, about two incompatible meanings of ξ in this program's own documents — and lists what is inherited from [2]. Appendix A describes the six check programmes and what each asserts; Appendix B records how this manuscript was drafted.

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2 Conventions, and the facts we use

2.1 The completed zeta function, and a warning

Throughout,

$$\Lambda(u) = \pi^{-u/2} \Gamma(u/2) \zeta(u), \quad \xi(u) = \frac{1}{2} u(u-1) \Lambda(u), \quad (2.1)$$

so that Λ satisfies $\Lambda(u) = \Lambda(1-u)$, is real on the real axis, and is holomorphic except for simple poles at $u = 0$ and $u = 1$ with residues -1 and $+1$; while ξ is entire and satisfies $\xi(u) = \xi(1-u)$. The trivial zeros of ζ cancel the poles of $\Gamma(u/2)$, so the zeros of Λ and of ξ are exactly the nontrivial zeros ρ of ζ .

Remark 2.1 (Two conventions in one program). The research notes on which Sections 3 and 6 are based [5, 6, 7] write ξ for what (2.1) calls Λ — that is, for the completed zeta *with* its two poles. The parent manuscript [2] uses (2.1), and so does this one. The two differ by exactly the rational factor R_ω of (4.1), which is the object of Section 4; the clash is harmless inside each document and lethal across them, and it is the reason this remark exists.

2.2 The shifted family

For $\omega > 0$ put

$$s = \frac{1}{2} + p, \quad a = \frac{1}{2} - \omega, \quad b = \frac{1}{2} + \omega, \quad c = -a = \omega - \frac{1}{2}, \quad (2.2)$$

so that $a + b = 1$, $b - a = 2\omega$ and $b - c = 1$, and let the transfer symbol and its Markov part be

$$K_\omega(p) = \frac{\xi(s - \omega)}{\xi(s + \omega)} = \frac{\xi(p + a)}{\xi(p + b)}, \quad \widetilde{K}_\omega(p) = \frac{\Lambda(p + a)}{\Lambda(p + b)}. \quad (2.3)$$

The *causal realization* of a symbol H holomorphic on $\operatorname{Re} p > \eta_0$ is the convolution with its inverse Laplace transform on any line $\operatorname{Re} p = \eta > \eta_0$, a distribution supported on $[0, \infty)$; we write V_ω for the causal realization of K_ω on $L^2(\mathbb{R})$, and, for $L > 0$ and P_L the multiplication by the indicator of $I_L = (-L/2, L/2)$,

$$V_{\omega,L} = P_L V_\omega P_L, \quad D_{\omega,L} = I - V_{\omega,L}^* V_{\omega,L} \quad (2.4)$$

for the compression to the window and its contraction defect. We write Q_L for the localized Weil form and m_L for its smallest eigenvalue on $L^2(I_L)$, the *margin*; Weil's criterion, used as input, is that the Riemann hypothesis holds if and only if $Q_L \geq 0$ for every L .

Finally we set

$$d = 2\omega, \quad m = d + 1 = 2\omega + 1, \quad \text{so} \quad a = \frac{2 - m}{2}, \quad b = \frac{m}{2}, \quad 2s = p + b, \quad (2.5)$$

which is the dictionary of [5, 6]: d is the dimension carried by every factor of $\widetilde{K}_\omega$ and $m = d + 1$ will turn out to be a rank. The family's range is $\omega \in (0, \frac{1}{2}]$, that is $m \in (1, 2]$.

2.3 What is inherited

The following are proved in the parent manuscript [2] and are used here as stated. We reprove nothing.

Proposition 2.2 (Unimodularity [2, Prop. 2.1]). $|K_\omega(i\tau)| = 1$ for every real τ with $\xi(\frac{1}{2} + \omega + i\tau) \neq 0$.

Proposition 2.3 (Flux and dichotomy [2, Prop. 2.3]). *For fixed ω , $L \mapsto \|V_{\omega,L}\|$ is nondecreasing with limit $\|V_{\omega}\|_{L^2(\mathbb{R})}$. If no zero of ξ satisfies $|\operatorname{Re} \rho - \frac{1}{2}| > \omega$, then K_{ω} is inner on the right half-plane, V_{ω} is unitary and causal on $L^2(\mathbb{R})$, and every $V_{\omega,L}$ is a contraction. If some zero satisfies $|\operatorname{Re} \rho - \frac{1}{2}| > \omega$, then $\sup_L \|V_{\omega,L}\| = \infty$.*

Proposition 2.4 (The first-order law [2, Sec. 2.2]). $\langle f, D_{\omega,L} f \rangle = 2\omega Q_L[f] + O(\omega^2)$ as $\omega \downarrow 0$, and $V_{0,L} = I$.

Proposition 2.5 (The pieces [2, Prop. 3.3]). $\widetilde{K}_{\omega} = K_{\omega}^{\Gamma} K_{\omega}^{\zeta}$ with

$$K_{\omega}^{\Gamma}(p) = \pi^{\omega} \frac{\Gamma(\frac{s-\omega}{2})}{\Gamma(\frac{s+\omega}{2})} = \int_0^{\infty} e^{-p\tau} k_{\omega}^{\Gamma}(\tau) d\tau, \quad k_{\omega}^{\Gamma}(\tau) = \frac{2\pi^{\omega}}{\Gamma(\omega)} (2 \sinh \tau)^{\omega-1} e^{\tau/2}, \quad (2.6)$$

$$K_{\omega}^{\zeta}(p) = \frac{\zeta(p+a)}{\zeta(p+b)} = \sum_{n \geq 1} \tilde{c}_n n^{-p}, \quad \tilde{c}_n = n^{\omega-\frac{1}{2}} \prod_{q|n} (1 - q^{-2\omega}), \quad (2.7)$$

the product over the primes q dividing n ; and $\tilde{c}_n > 0$ for every $n \geq 1$ and every real $\omega > 0$.

2.4 Complete monotonicity and Blaschke factors

A function H on (p_0, ∞) is *completely monotone* if $(-1)^k H^{(k)} \geq 0$ for every $k \geq 0$; by Bernstein's theorem this holds if and only if $H(p) = \int_{[0,\infty)} e^{-pt} d\mu(t)$ for a positive measure μ whose integral converges for $p > p_0$, and μ is then the causal inverse Laplace kernel of H [12]. Products of completely monotone functions are completely monotone. A *Blaschke factor* of the right half-plane is

$$B_{\gamma}(p) = \frac{p - \gamma}{p + \gamma}, \quad \gamma > 0, \quad (2.8)$$

holomorphic and bounded by 1 on $\operatorname{Re} p > 0$, unimodular on the axis, with causal kernel $\delta_0 - 2\gamma e^{-\gamma t} \mathbf{1}_{t>0}$. A finite product of Blaschke factors is *inner*: $|\cdot| \leq 1$ throughout the open right half-plane. We will use repeatedly that for $\beta > \alpha$ the Beta integral gives

$$\frac{\Gamma(z + \alpha)}{\Gamma(z + \beta)} = \frac{1}{\Gamma(\beta - \alpha)} \int_0^{\infty} e^{-zt} e^{-\alpha t} (1 - e^{-t})^{\beta - \alpha - 1} dt, \quad (2.9)$$

a completely monotone function of z on $z > -\alpha$, with a manifestly positive density; and that complete monotonicity is preserved by an increasing affine change of the variable.

3 The transfer is an invariant of the rank

The dictionary of [5, 6] makes every factor of $\widetilde{K}_{\omega}$ a d -dimensional formula: the archimedean factor is the Riesz integral over $\mathbb{R}^d$, the comb's weights are the Jordan totient J_d , and the residue of $\widetilde{K}_{\omega}$ at $p = b$ is $|S^d|/2\zeta(d+1)$, the volume term of a lattice of rank $d+1$. This section says what the object is at integer d , and the answer fixes the meaning of m for the rest of the manuscript.

Proposition 3.1 (Primitive Epstein scattering). *Let $L \subset \mathbb{R}^m$ be a lattice of covolume 1, L^* its dual, and for $\operatorname{Re} s > m/2$*

$$Z_L(s) = \sum_{0 \neq v \in L} |v|^{-2s}, \quad E_L^*(s) = \sum_{v \in L \text{ primitive}} |v|^{-2s}.$$

Then $Z_L = \zeta(2s) E_L^$, both continue meromorphically, and*

$$\Lambda(2s) E_L^*(s) = \Lambda(m - 2s) E_{L^*}^*(m/2 - s). \quad (3.1)$$

In particular the reflection ratio $E_L^*(s)/E_{L^*}^*(m/2 - s) = \Lambda(m - 2s)/\Lambda(2s)$ is independent of L : it is an invariant of the rank m alone.

Proof. Every nonzero $v \in L$ is uniquely kv_0 with $k \geq 1$ and v_0 primitive, so $Z_L(s) = \sum_{k \geq 1} k^{-2s} E_L^*(s) = \zeta(2s) E_L^*(s)$. Poisson summation in dimension m gives $\theta_L(1/x) = x^{m/2} \theta_{L^*}(x)$ for $\theta_L(x) = \sum_{v \in L} e^{-\pi x |v|^2}$ at covolume one, whence Riemann's theta argument yields $\pi^{-s} \Gamma(s) Z_L(s) = \pi^{-(m/2-s)} \Gamma(m/2 - s) Z_{L^*}(m/2 - s)$. Divide by the factorization and use $\pi^{-s} \Gamma(s) \zeta(2s) = \Lambda(2s)$ and $\pi^{-(m/2-s)} \Gamma(m/2 - s) \zeta(m - 2s) = \Lambda(m - 2s)$. See also [14, Ch. 4]. $\square$

Proposition 3.2 (The integer points of the family). *With the dictionary (2.5), for every real $d > 0$*

$$\widetilde{K}_\omega(p) = \frac{\Lambda(p + a)}{\Lambda(p + b)} = \frac{\Lambda(2s - d)}{\Lambda(2s)} = \frac{\Lambda(m - 2s)}{\Lambda(2s)}, \quad (3.2)$$

and for every integer $d \geq 1$ this equals $E_L^*(s)/E_{L^*}^*(m/2 - s)$ for every covolume-one lattice L of rank $m = d + 1$. Moreover $s(-p) = \frac{m}{2} - s(p)$ identically, so the transfer's reflection $p \mapsto -p$ is the Epstein functional equation $s \mapsto \frac{m}{2} - s$; and on $\operatorname{Re} p = 0$ the two arguments are complex conjugates, so $|\widetilde{K}_\omega| = 1$ there.

Proof. $2s = p + b$ gives $2s - d = p + b - 2\omega = p + a$ and $m - 2s = (2\omega + 1) - (p + b) = b - p$, and $\Lambda(w) = \Lambda(1 - w)$ turns $\Lambda(b - p)$ into $\Lambda(1 - b + p) = \Lambda(p + a)$ since $1 - b = a$; this is (3.2), and the identification at integer d is Proposition 3.1. For the reflection, $s(p) = \frac{p+b}{2}$ and $\frac{m}{2} - s(p) = \frac{(2\omega+1)-p-b}{2} = \frac{b-p}{2} = s(-p)$. On $p = i\tau$ this is $\overline{s(p)}$, and E^* has real coefficients, so for a self-dual L the ratio is $E^*(s)/\overline{E^*(s)}$. $\square$

Corollary 3.3 (One object, three entries). *At integer d the three entries of the dictionary are the three parts of one constant term. The archimedean factor $\pi^{d/2} \Gamma(s - \frac{d}{2}) / \Gamma(s) = \int_{\mathbb{R}^d} |t + i|^{-(p+b)} dt$ is the integral over a d -dimensional unipotent radical [4]; the comb $\zeta(2s - d) / \zeta(2s) = \sum_n J_d(n) n^{-2s}$ is the count of that radical modulo n , which is what the Jordan totient $J_d(n) = n^d \prod_{q|n} (1 - q^{-d})$ counts; and the residue of E^* at its rightmost pole $s = \frac{m}{2}$ is $|S^d| / 2\zeta(d + 1)$, the sphere of $\mathbb{R}^{d+1}$ times the primitive density in $\mathbb{Z}^{d+1}$, which is $\operatorname{Res}_{p=b} \widetilde{K}_\omega$ [6]. In standard language $g \mapsto E_{\mathbb{Z}^m}^* g(s)$ is the degenerate Eisenstein series of the maximal parabolic $P_{1,m-1}$ of $SL(m)$ and $\Lambda(m - 2s) / \Lambda(2s)$ is its c -function [14]; at $m = 2$ it is the classical $\varphi(s) = \Lambda(2s - 1) / \Lambda(2s)$ of $PSL(2, \mathbb{Z}) \backslash \mathbb{H}$ [15, 16].*

Remark 3.4 (The reading, and its limit). So $d = 1, 2, 3, \dots$ are all realized, as ranks $2, 3, 4, \dots$: the transfer is the scattering matrix of the space of unimodular lattices of rank $d + 1$, at every integer and not only at $d = 1$. The family's range contains exactly one of them, $d = 1$. What Proposition 3.1 does not give is any *individual* lattice: the ratio is the same for all of them, so no property of a particular L — its theta series, its minimum, its positivity — is recoverable from $\widetilde{K}_\omega$. That will matter in Section 7. The group-theoretic sentence of Corollary 3.3 is the standard interpretation of the verified formulas and is not something proved here.

4 The wall at rank two

The family's range is $m \in (1, 2]$, the first gap of the rank sequence. This section shows that its upper endpoint is not a convention but the zero of $a = \frac{2-m}{2}$, and that three distinct things happen there. Two of them are in this section; the third, which is what the endpoint costs, is Section 5.

4.1 The factorization, and where the two poles come from

Proposition 4.1 (Factorization [2, Prop. 3.1]). $K_\omega = R_\omega \widetilde{K}_\omega$ with

$$R_\omega(p) = \frac{(s - \omega)(s - \omega - 1)}{(s + \omega)(s + \omega - 1)} = \frac{(p + a)(p - b)}{(p + b)(p - a)} = 1 - \frac{4\omega b}{p + b} - \frac{4\omega a}{p - a}. \quad (4.1)$$

Apart from its poles at the zeros of $\Lambda(p + b)$ and its zeros at the zeros of $\Lambda(p + a)$, $\widetilde{K}_\omega$ has exactly two poles, at $p = b$ and $p = -a = c$, and exactly two zeros, at $p = a$ and $p = -b$; R_ω cancels all four. Both factors are unimodular on $\text{Re } p = 0$.

The two cancelled poles are the poles of $\Lambda(p + a)$, at $p + a = 1$ and $p + a = 0$. They have different origins, and the distinction is what makes the repair of Section 4.2 the natural one.

Table 1: The two poles of $\widetilde{K}_\omega$ in the right half-plane, by origin. Writing $\Lambda(p + a) = \pi^{-(p+a)/2} \Gamma(\frac{p+a}{2}) \zeta(p + a)$, the pole at $p = b$ comes from ζ and the pole at $p = c$ from Γ . The remaining poles of $\Gamma(\frac{p+a}{2})$, at $p + a = -2, -4, \dots$, are cancelled by the trivial zeros of $\zeta(p + a)$, and $\zeta(0) = -\frac{1}{2} \neq 0$, so the pole at $p + a = 0$ is exactly the archimedean one.

pole	origin	in $\text{Re } p > 0$ iff
$p = b = \frac{m}{2}$	$\zeta(p + a)$ at $p + a = 1$: the <i>comb</i>	always
$p = c = \frac{m-2}{2}$	$\Gamma(\frac{p+a}{2})$ at $p + a = 0$: the <i>archimedean factor</i>	$m > 2$

Proposition 4.2 (The second factor is a Blaschke factor exactly past the wall). *For every $\omega > 0$,*

$$R_\omega(p) = B_b(p) \cdot \frac{p + a}{p - a}, \quad \text{and} \quad \frac{p + a}{p - a} = B_c(p) \quad \text{with } c = \omega - \frac{1}{2}. \quad (4.2)$$

Consequently

- for $\omega < \frac{1}{2}$ one has $c < 0$, and $R_\omega = B_b/B_a$ has a pole at $p = a > 0$: it is unimodular on the axis but not inner, and $\sup_{\text{Re } p > 0} |R_\omega| = \infty$;
- for $\omega \geq \frac{1}{2}$ one has $c \geq 0$, and $R_\omega = B_b B_c$ is a product of two Blaschke factors of the right half-plane, hence inner: $|R_\omega| \leq 1$ on $\text{Re } p > 0$.

At $\omega = \frac{1}{2}$ exactly, $c = 0$ and $B_c \equiv 1$: the two regimes meet at the single factor B_b .

Proof. Equation (4.1) and $a - 1 = -b$, $b - 1 = -a$ give the first identity; and $\frac{p+a}{p-a} = \frac{p-(-a)}{p+(-a)} = B_{-a} = B_c$. A Blaschke factor with positive parameter is inner and a product of inner functions is inner; for $\omega < \frac{1}{2}$ the pole at $p = a > 0$ lies in the open right half-plane. $\square$

4.2 Complete monotonicity survives the crossing

The parent manuscript proves that $\widetilde{K}_\omega$ is completely monotone on (b, ∞) [2, Prop. 3.2] and that $\widetilde{K}_\omega = \frac{p+a}{p-a} \widehat{K}_\omega$ is too [2, Prop. 3.6], so that $K_\omega = B_b \widetilde{K}_\omega$ exhibits the transfer as a positive measure minus twice its exponential moving average. The first of these carries no restriction on ω ; the second, as proved there, does. We record both facts and repair the second.

Proposition 4.3 (Complete monotonicity, uncapped). *For every real $\omega > 0$:*

- (a) $\widetilde{K}_\omega$ is completely monotone on (b, ∞) ;

(b) $\widehat{K}_\omega(p) = \frac{p+a}{p-a} \widetilde{K}_\omega(p)$ is completely monotone on (b, ∞) .

For $\omega \leq \frac{1}{2}$ part (b) is [2, Prop. 3.6]. For $\omega \geq \frac{1}{2}$ its proof there fails and the following grouping replaces it: with $z = \frac{p+a}{2}$ and $\widehat{K}_\omega = \widehat{K}_\omega^\Gamma K_\omega^\zeta$,

$$\widehat{K}_\omega^\Gamma(p) = \pi^\omega \frac{\Gamma(z+c)}{\Gamma(z+\omega)} \cdot \frac{\Gamma(z+1)}{\Gamma(z+c+1)}, \quad (4.3)$$

a product of two Gamma ratios whose parameter gaps are $\frac{1}{2}$ and c .

Proof. (a) is [2, Prop. 3.2]; its proof uses only that $p > b$ implies $p+a > 1$, which holds for every $\omega > 0$, and is therefore uncapped.

For (b) with $\omega \leq \frac{1}{2}$, the parent writes $\frac{p+a}{p-a} = 1 + \frac{2a}{p-a}$, whose causal kernel is $\delta_0 + 2a e^{at} \mathbf{1}_{t>0}$; this is a positive measure if and only if $a \geq 0$, that is if and only if $\omega \leq \frac{1}{2}$. Past the wall the same expression has $2a < 0$ and the argument is unavailable.

For $\omega \geq \frac{1}{2}$ regroup instead. From $(p+a)\Gamma(\frac{p+a}{2}) = 2\Gamma(z+1)$, from $\frac{p+b}{2} = z+\omega$, and from $p-a = 2(z+c)$,

$$\frac{p+a}{p-a} \cdot \pi^\omega \frac{\Gamma(\frac{p+a}{2})}{\Gamma(\frac{p+b}{2})} = \pi^\omega \frac{\Gamma(z+1)}{(z+c)\Gamma(z+\omega)} = \pi^\omega \frac{\Gamma(z+c)}{\Gamma(z+\omega)} \cdot \frac{\Gamma(z+1)}{\Gamma(z+c+1)},$$

the last step by $\Gamma(z+c+1) = (z+c)\Gamma(z+c)$. The parameter gaps are $\omega - c = \frac{1}{2} > 0$ and $(c+1) - 1 = c \geq 0$, so each factor is completely monotone in z by (2.9) (the second is identically 1 when $c = 0$), and $z = \frac{p+a}{2}$ is an increasing affine function of p with $z > \frac{1}{2}$ on $p > b$, inside both domains. The comb K_ω^ζ is completely monotone on (b, ∞) with the positive coefficients (2.7). Products of completely monotone functions are completely monotone. $\square$

Remark 4.4 (The two groupings are exchanged at the wall). Proposition 4.3 proves the same statement twice, by arguments that are available on complementary ranges and meet where both degenerate. At $\omega = \frac{1}{2}$ the parent's factor $\frac{p+a}{p-a}$ is identically 1 and so is the second Gamma ratio of (4.3). At $\omega = 1$ the two Gamma ratios telescope: $c = \frac{1}{2}$, $\omega = 1$, and $\widehat{K}_1^\Gamma(p) = \pi/(z+\frac{1}{2}) = 2\pi/(p+\frac{1}{2})$, whose kernel is $2\pi e^{-\tau/2}$.

Remark 4.5 (The trade). Table 2 collects what changes at $m = 2$ and what does not. The factor $\frac{p+a}{p-a}$ does not *lose* a positivity there: it *exchanges* complete monotonicity for innerness. Everything else in the decomposition is positive on both sides. Whether that exchange is the same fact as the failure of the lattice on the same interval is the question of Section 7, and the answer is no.

Table 2: What the wall separates. Only the first row changes, and it changes by exchange rather than by loss.

object	$m < 2$	$m > 2$
$\frac{p+a}{p-a}$	completely monotone, <i>not</i> inner	inner, <i>not</i> completely monotone
$R_\omega = B_b \cdot \frac{p+a}{p-a}$	not inner (pole at $p = a > 0$)	inner
$\widetilde{K}_\omega$	completely monotone	completely monotone
$\widehat{K}_\omega$	completely monotone (parent's grouping)	completely monotone (regrouped)
$k_\omega^\Gamma, \widetilde{c}_n$	positive	positive

5 What the criterion is worth

The third thing that happens at $a = 0$ is that the poles of K_ω leave the closed right half-plane. That costs the criterion everything.

5.1 Vacuity past the wall

Proposition 5.1 (Vacuity). *Let $\omega \geq \frac{1}{2}$. Then no nontrivial zero of ζ satisfies $|\operatorname{Re} \rho - \frac{1}{2}| > \omega$. Hence, by Proposition 2.3, K_ω is inner on $\operatorname{Re} p > 0$, V_ω is unitary and causal on $L^2(\mathbb{R})$, and*

$$\|V_{\omega,L}\| \leq 1 \quad \text{for every } L > 0, \quad (5.1)$$

unconditionally. For $\omega > \frac{1}{2}$ the only arithmetic input is $\zeta(u) \neq 0$ for $\operatorname{Re} u > 1$ — the Euler product — together with the functional equation, which give $0 \leq \operatorname{Re} \rho \leq 1$.

Proof. $0 \leq \operatorname{Re} \rho \leq 1$ gives $|\operatorname{Re} \rho - \frac{1}{2}| \leq \frac{1}{2} \leq \omega$, so the hypothesis of Proposition 2.3 holds with nothing assumed. Equivalently at the level of poles: the poles of K_ω lie at $p = \rho - b$, with $\operatorname{Re} p \leq 1 - b = a \leq 0$, and $a < 0$ as soon as $\omega > \frac{1}{2}$. $\square$

Remark 5.2 (The boundary point). At $\omega = \frac{1}{2}$ exactly, $a = 0$ and $\operatorname{Re} p = \operatorname{Re} \rho - 1$ is negative for every zero but has supremum 0, the zeros approaching $\operatorname{Re} \rho = 1$ as the height grows. The hypothesis of Proposition 2.3 is still satisfied by the Euler product, since it is a strict inequality; moving the realization contour to the axis there uses $\zeta(1 + it) \neq 0$, the classical zero-free region [17], and not the Euler product. Nothing below turns on the boundary point: the informative range is the open interval $(0, \frac{1}{2})$ either way.

Corollary 5.3 (Where the criterion has content). *By Proposition 2.3 in both directions, $\sup_L \|V_{\omega,L}\| \leq 1$ if and only if every zero satisfies $|\operatorname{Re} \rho - \frac{1}{2}| \leq \omega$. Therefore*

- for $\omega \geq \frac{1}{2}$ the statement is unconditionally true and is equivalent to nothing about the zeros: the criterion is vacuous;
- for $\omega \in (0, \frac{1}{2})$ it is equivalent to the zero-free strip of half-width ω , a nontrivial statement;
- the Riemann hypothesis is the conjunction over all $\omega \in (0, \frac{1}{2})$, that is the limit $\omega \downarrow 0$.

5.2 Which endpoint is the Riemann hypothesis

Corollary 5.3 makes ω exactly the half-width of the zero-free strip the criterion certifies, so the criterion is *weakest* at the right endpoint of the family and *sharpest* at the left. That inverts a natural first reading, in which rank two — the modular surface, the lattice ζ itself comes from — looks like where the Riemann hypothesis lives. It is where the *object* lives; it is not where the criterion has strength.

The bottom row deserves a sentence of its own. At rank one the primitive Epstein zeta of $\mathbb{Z} \subset \mathbb{R}$ is the constant 2 (only ± 1 is primitive), its reflection ratio is 1, and the transfer degenerates to the identity; and by Proposition 2.4 the first-order defect of the compression is the localized Weil form. *The Riemann hypothesis is the derivative of the rank-(d+1) scattering matrix at rank one; the Euler product is its value at rank two; and the gap between the two smallest ranks is filled by the zero-free strips.*

Table 3: The family runs from the Euler product to the Riemann hypothesis, and the direction is from the larger rank to the smaller.

m	the object	the criterion
$m = 2, \omega = \frac{1}{2}$	rank two: $\mathbb{Z}^2$, the modular surface, $\varphi(s) = \frac{\Lambda(2s-1)}{\Lambda(2s)}$	its <i>value</i> is the Euler product
$m \in (1, 2)$	no lattice (Section 6)	zero-free strips of half-width $\frac{m-1}{2}$
$m = 1, \omega = 0$	rank one: $E_{\mathbb{Z}}^* \equiv 2, \tilde{K}_0 = 1, V_0 = I$	its <i>derivative</i> is the Weil form

5.3 The instrument, and a calibration

The parent manuscript’s registered assembly [2, Sec. 5] computes $\|V_{\omega,L}\|$ and $\lambda_{\min}(D_{\omega,L})$ from Γ , $\sinh$, $\exp$ and the integers below e^L . It continues past the wall with one change, and the change is arithmetic rather than analytic.

Remark 5.4 (The assembly needs no second Blaschke factor). The assembly is built from the *partial fractions* of R_{ω} in (4.1), namely $\kappa_{\omega} = k_{\omega}^{\Gamma} - 4\omega b J_{-b} - 4\omega a J_a$ with $J_{\gamma}(\tau) = \int_0^{\tau} e^{\gamma(\tau-u)} k_{\omega}^{\Gamma}(u) du$, which is the same algebra at every ω , with $a < 0$ merely making the second smoothing decay instead of grow. The Blaschke factorization of Proposition 4.2 enters the proofs of positivity, not the computation. The one obstacle is that the parent programme writes the $k = 0$ Jacobi recurrence coefficient of its Gauss–Jacobi rule as $(\beta^2 - \alpha^2)/(\delta(\delta + 2))$ with $\delta = \alpha + \beta$, which is $0/0$ when $\alpha = \beta = 0$, that is at $\omega = 1$ exactly and nowhere else; its removable value is $(\beta - \alpha)/(\alpha + \beta + 2)$. With that one line changed the assembly’s own certificate,

$$\int_0^{\infty} e^{-p\tau} \kappa_{\omega}(\tau) d\tau = \pi^{\omega} \frac{\Gamma(\frac{a+p}{2})}{\Gamma(\frac{b+p}{2})} \cdot \frac{(p+a)(p-b)}{(p+b)(p-a)}$$

[2, Prop. 5.3], holds at $\omega = 1, \frac{3}{2}, 2$ to 3×10^{-13} (Appendix A).

By Proposition 5.1 the contraction data past the wall is known in advance, so running the assembly there is a *calibration* and not a test. Table 4 records it.

Table 4: Calibration at $L = \log 3$, $N = 24$ sine modes, 32 Gauss–Jacobi nodes per atom; $m_L^{(24)} = 6.247514 \times 10^{-8}$. Contraction here is Proposition 5.1, not these numbers. What the table shows is that the instrument returns it, and that the first-order law $\lambda_{\min}(D_N)/2\omega = m_L^{(N)}$ degrades smoothly as ω grows, with no feature at the wall.

ω	rank m	mass on window	$1 - \ V_N\ $	$\lambda_{\min}(D_N)$	$\lambda_{\min}(D_N)/(2\omega m_L^{(24)})$
0.1	1.2	0.940662658	6.61×10^{-9}	1.3217×10^{-8}	1.058
$\frac{1}{2}$	2	0.774030153	4.58×10^{-8}	9.1686×10^{-8}	1.468
1	3	0.707371247	1.72×10^{-7}	3.4445×10^{-7}	2.757
$\frac{3}{2}$	4	0.762741187	5.38×10^{-7}	1.0761×10^{-6}	5.741
2	5	0.891128215	1.55×10^{-6}	3.1068×10^{-6}	12.432

Remark 5.5 (Which direction these numbers certify). $\lambda_{\min}(D_N)$ is a trial-space Rayleigh quotient on the N -mode sine basis, and for f in that space $\langle f, D_N f \rangle = \|f\|^2 - \|P_N V f\|^2 \geq \|f\|^2 - \|V f\|^2$, so $\lambda_{\min}(D_N) \geq \lambda_{\min}(D_{\omega,L})$: it *overstates* the margin, and a positive computed value does *not* certify contraction. Every recorded margin in this program is one-sided in this direction. The wall is invisible in Table 4 — as it must be, since it is not a wall in the transfer but a wall in what the transfer is evidence for.

6 Inside the gap there is no lattice

Section 3 put a rank- m lattice at every integer $m \geq 2$ and the family in the first gap $m \in (1, 2]$. This section shows that the lattice does not interpolate into the gap, and identifies exactly what stops it.

6.1 The interpolation exists and is forced

Write $q = e^{-\pi t}$ and $\theta = \sum_{k \in \mathbb{Z}} q^{k^2} = 1 + u$ with $u = 2q + 2q^4 + 2q^9 + \dots$, and for $j \geq 1$ let $R_j(n)$ be the number of ordered j -tuples of positive squares summing to n , so that $u^j = \sum_n 2^j R_j(n) q^n$. Let

$$s(n) = \min\{j : R_j(n) > 0\} \quad (6.1)$$

be the least number of positive squares summing to n .

Proposition 6.1 (The interpolation [7, Prop. 3.1]). *θ does not vanish on the upper half-plane, so $\theta^m = \exp(m \log \theta)$ is well defined there for every real m , and*

$$\theta^m = \sum_{j \geq 0} \binom{m}{j} u^j, \quad r_m(n) = \sum_{j=1}^n \binom{m}{j} 2^j R_j(n), \quad (6.2)$$

the sum terminating because $R_j(n) = 0$ for $j > n$. Each $r_m(n)$ is a polynomial in m of degree at most n which at every integer $m \geq 0$ is the representation number of $\mathbb{Z}^m$; being of degree $\leq n$ and matching at $m = 0, \dots, n$, it is the unique such interpolation.

6.2 The sign law, the free region, and the forcing

Proposition 6.2 (The sign law). *Let $k \geq 0$ be an integer and $m \in (k, k+1)$. Then*

$$\text{sign} \binom{m}{j} = (-1)^{\max(0, j-k-1)}, \quad (6.3)$$

that is: $\binom{m}{j} > 0$ for $1 \leq j \leq k+1$, $\binom{m}{k+2} < 0$, and the sign alternates thereafter. No $\binom{m}{j}$ vanishes.

Proof. $\binom{m}{j} = \frac{1}{j!} \prod_{i=0}^{j-1} (m-i)$, and for integer i one has $m-i < 0$ exactly when $i \geq k+1$; among $i = 0, \dots, j-1$ the number of such i is $\max(0, j-1-k)$. None vanishes because m is not an integer. $\square$

Proposition 6.3 (The free region). *For $m \in (k, k+1)$ and $1 \leq n \leq k+1$, $r_m(n) > 0$.*

Proof. $R_j(n) = 0$ for $j > n$, so every contributing j satisfies $j \leq n \leq k+1$ and has $\binom{m}{j} > 0$ by Proposition 6.2; and $R_n(n) = 1$, so the sum is not empty. $\square$

So negativity can begin only at $n \geq k+2$, and the first place at which it is forced — where no positive term is present at all — is governed by an arithmetic quantity.

Proposition 6.4 (Where the forcing lives, and where it stops). *Put $N_k = \min\{n \geq 1 : s(n) \geq k+2\}$. Then*

$$N_0 = 2, \quad N_1 = 3, \quad N_2 = 7, \quad (6.4)$$

with $s(N_k) = k+2$ in each case; and N_k does not exist for any $k \geq 3$, because Lagrange's four-square theorem [13, Thm. 369] gives $s(n) \leq 4$ for every $n \geq 1$.

Proof. The three values are a finite verification. 2 is not a square and $2 = 1 + 1$, so $s(2) = 2$. 3 is not a square, and $3 - 1 = 2$ is not a square, so 3 is not a sum of two positive squares; $3 = 1 + 1 + 1$, so $s(3) = 3$. 7 is not a square; $7 - 1 = 6$ and $7 - 4 = 3$ are not squares, so 7 is not a sum of two positive squares; $7 - 1 - 1 = 5$, $7 - 1 - 4 = 2$ and $7 - 4 - 4 < 0$ are not squares, so 7 is not a sum of three positive squares; and $7 = 4 + 1 + 1 + 1$, so $s(7) = 4$. For the last clause, Lagrange writes every n as a sum of four squares; discarding the zero entries leaves at most four positive ones. $\square$

Proposition 6.5 (The forced coefficient). *For $k = 0, 1, 2$ and every $m \in (k, k + 1)$,*

$$r_m(N_k) < 0, \quad \text{while} \quad r_m(n) > 0 \quad \text{for every } 1 \leq n < N_k. \quad (6.5)$$

Explicitly $r_m(2) = 2m(m - 1)$, $r_m(3) = \frac{4}{3}m(m - 1)(m - 2)$ and $r_m(7) = 64\binom{m}{4} + 128\binom{m}{7}$.

Proof. At $n = N_k$ every contributing j has $R_j(N_k) > 0$, hence $j \geq s(N_k) = k + 2$, so every term carries a binomial of the signs of Proposition 6.2 at or beyond $j = k + 2$.

For $k = 0$ and $k = 1$ only one j contributes at all: $R_j(2) = 0$ except $R_2(2) = 1$, and $R_j(3) = 0$ except $R_3(3) = 1$. Hence $r_m(2) = 4\binom{m}{2}$ and $r_m(3) = 8\binom{m}{3}$, both negative on their gap.

For $k = 2$, $R_j(7) = 0$ except $R_4(7) = 4$ and $R_7(7) = 1$, so $r_m(7) = 64\binom{m}{4} + 128\binom{m}{7}$ and

$$r_m(7) = m(m - 1)(m - 2)(m - 3) \left[\frac{8}{3} + \frac{8}{315}(m - 4)(m - 5)(m - 6) \right].$$

On $(2, 3)$ the prefactor is negative, and $(m - 4)(m - 5)(m - 6) \in (-24, -6)$, so the bracket lies in $(\frac{8}{3} - \frac{192}{315}, \frac{8}{3} - \frac{48}{315}) \subset (2.05, 2.52)$ and is positive; hence $r_m(7) < 0$.

For the second clause, $n \leq k + 1$ is Proposition 6.3, which settles $n = 1$ for $k = 0$, $n \leq 2$ for $k = 1$ and $n \leq 3$ for $k = 2$. Three cases remain, all on $(2, 3)$. From $R_1(4) = R_4(4) = 1$, $r_m(4) = 2m + 16\binom{m}{4} = 2m[1 + \frac{1}{3}(m - 1)(m - 2)(m - 3)]$, and $(m - 1)(m - 2)(m - 3) \in (-2, 0)$ there, so the bracket exceeds $\frac{1}{3}$. From $R_2(5) = 2$ and $R_5(5) = 1$, $r_m(5) = 8\binom{m}{2} + 32\binom{m}{5}$, and on $(2, 3)$ both binomials are positive by Proposition 6.2. From $R_3(6) = 3$ and $R_6(6) = 1$, $r_m(6) = 24\binom{m}{3} + 64\binom{m}{6}$ with $\binom{m}{3} > 0$ and $\binom{m}{6} < 0$, and $\frac{64\binom{m}{6}}{24\binom{m}{3}} = \frac{8}{3} \cdot \frac{(m-3)(m-4)(m-5)}{120}$, whose modulus is below $\frac{8}{3} \cdot \frac{6}{120} = \frac{2}{15} < 1$. $\square$

Corollary 6.6 (No lattice in the gap). *For every $m \in (1, 2)$ — that is, every d strictly inside the family's range — $\theta(it)^m = \sum_n r_m(n)e^{-\pi n t}$ is not completely monotone: being a Laplace expansion with discrete support, its representing measure is forced by uniqueness to be $\sum_n r_m(n)\delta_{\pi n}$, and $r_m(3) < 0$. There is no positive measure and hence no point set.*

Remark 6.7 (What the mechanism is, and what it is not). Propositions 6.2–6.5 say that the point count's failure is a statement about *sums of squares*: on the gap $(k, k + 1)$ it is forced by the least integer needing $k + 2$ positive squares, and it stops being forced at $k = 3$ for one reason only, Lagrange's theorem. The scan reported in [7] found the first negative coefficient at $n = 2$ on $(0, 1)$, $n = 3$ on $(1, 2)$, $n = 7$ on $(2, 3)$, at varying n on $(3, 4)$, and none at all for $m \geq 4$; the first three of those are now Proposition 6.5 and the last is Proposition 6.4.

What is *not* covered is negativity produced by competition rather than by forcing. On $(3, 4)$ nothing is forced, yet negative coefficients occur: there the $j = k + 2$ terms compete with positive ones instead of standing alone. Two statements are therefore recorded as conjectures and are not used anywhere in this manuscript: that $r_m(n) \geq 0$ for every n and every real $m \geq 4$, and that the negative coefficients on $(3, 4)$ arise from competition. Both are questions about sums of squares rather than about ζ .

7 Exponent versus binomial index

Two positivity statements now sit on the same one-parameter family. The transfer's rational factor exchanges complete monotonicity for innerness at $m = 2$ (Remark 4.5); the point count loses complete monotonicity on $m \in (1, 2)$ (Corollary 6.6). The intervals look the same. This section shows that they are not the same statement, that the agreement is forced by arithmetic of a very cheap kind, and what does separate the two sides.

7.1 The transfer has exactly one positivity threshold

Proposition 7.1 (One threshold, at $m = 2$). *In the decomposition $K_\omega = R_\omega K_\omega^\Gamma K_\omega^\zeta$:*

- $k_\omega^\Gamma(\tau) > 0$ for every $\tau > 0$ and every real $\omega > 0$;
- $\tilde{c}_n > 0$ for every $n \geq 1$ and every real $\omega > 0$, and likewise $J_d(n) > 0$ for every real $d > 0$;
- $\widehat{K}_\omega$ and $\widehat{K}_\omega$ are completely monotone on (b, ∞) for every real $\omega > 0$ (Proposition 4.3);
- and the only object in the decomposition whose positivity depends on ω at all is $\frac{p+a}{p-a}$, of causal kernel $\delta_0 + 2a e^{at} \mathbf{1}_{t>0}$, positive exactly when $a \geq 0$, that is when $m \leq 2$.

Hence the transfer has exactly one positivity threshold in m , and it is at $m = 2$.

Proof. $\Gamma(\omega) > 0$ and $\sinh \tau > 0$ for $\omega, \tau > 0$, which gives the first item from (2.6); each local factor $1 - q^{-2\omega}$ and $1 - q^{-d}$ lies in $(0, 1)$, which gives the second from (2.7) and Corollary 3.3; the third is Proposition 4.3; the fourth is the causal kernel of $1 + \frac{2a}{p-a}$. $\square$

Remark 7.2 (The one other feature at an integer, and it is the wrong kind). At $m = 3$, that is $\omega = 1$, the archimedean exponent $\omega - 1 = \frac{m-3}{2}$ vanishes and k_ω^Γ passes from unbounded at the origin ($m < 3$) through bounded ($m = 3$, where it is exactly $2\pi e^{\tau/2}$) to vanishing there ($m > 3$). That is a *regularity* threshold, not a positivity one — k_ω^Γ is positive throughout — and it points the wrong way: at $m = 3$ the point count *regains* complete monotonicity, while nothing on the transfer side is regained or lost.

7.2 The two positivity sets diverge

Table 5: The two positivity statements on $m > 1$. They agree on $(1, 2)$ and differ on $(2, 3)$; and at $m = 3$ the point count is completely monotone while the transfer's rational factor is not.

m	$\theta(it)^m$ compl. mon.	$\frac{p+a}{p-a}$ compl. mon.	criterion has content
(1, 2)	no — $r_m(3) < 0$	yes ($a > 0$)	yes
2	yes (rank-two lattice)	yes ($a = 0$, factor $\equiv 1$)	no (vacuous)
(2, 3)	no — $r_m(7) < 0$	no ($a < 0$)	no (vacuous)
3	yes (rank-three lattice)	no ($a < 0$)	no (vacuous)

Corollary 7.3 (No equivalence). *The point count's failure set contains $(2, 3)$, where the criterion is already vacuous by Proposition 5.1; and at $m = 3$ the point count is completely monotone while $\frac{p+a}{p-a}$ is not. Hence no statement of the form “ $\theta(it)^m$ is completely monotone if and only if [any positivity of the transfer]” can hold.*

Proof. Propositions 6.5 and 5.1 give the first clause and Propositions 7.1 and 6.1 the second; at $m = 3$ the coefficients $r_3(n)$ are representation numbers of $\mathbb{Z}^3$ and are nonnegative, while $a = -\frac{1}{2} < 0$. $\square$

Remark 7.4 (The agreement is about the number two). The criterion's window is bounded below by $m = 1$, where $\omega = 0$ and the transfer degenerates to the identity, and above by $m = 2$, where a changes sign. Those are the two smallest ranks, and the first gap of the rank sequence is $(1, 2)$ by definition. Any pair of thresholds sitting at the two smallest ranks produces the first gap. The coincidence of intervals is therefore a statement about the number 2 and not about either positivity, and it is not evidence of a common mechanism.

7.3 Where the rank sits

What does separate the two sides is the *position* the rank occupies in each.

Proposition 7.5 (Exponent versus binomial index). *Every positivity in the transfer is produced by one of two mechanisms, in each of which the rank enters as an exponent of a positive quantity.*

- **Beta densities.** For $\beta > \alpha$, (2.9) represents $\Gamma(z + \alpha)/\Gamma(z + \beta)$ by the density $\Gamma(\beta - \alpha)^{-1} e^{-\alpha t} (1 - e^{-t})^{\beta - \alpha - 1}$, positive for every real $\beta - \alpha > 0$. This is the archimedean factor $k_\omega^\Gamma \propto (2 \sinh \tau)^{\omega - 1} e^{\tau/2}$ and both Gamma ratios of the regrouping (4.3).
- **Euler factors.** The comb's local factor at a prime q is

$$\frac{1 - q^{-u}}{1 - q^{-(u-d)}} = 1 + \sum_{k \geq 1} q^{(k-1)d} (q^d - 1) q^{-ku}, \quad (7.1)$$

whose coefficients are positive for every real $d > 0$ because $q^d > 1$; multiplying over $q \mid n$ recovers $J_d(n) = n^d \prod_{q \mid n} (1 - q^{-d})$.

In both, positivity is a local inequality — at one delay τ , at one prime q — in which d appears as an exponent, and an exponent of a positive quantity is positive at every real value. The point count has neither structure: by Proposition 6.1 its unique continuation is a binomial series in which the rank appears as an index, and by Proposition 6.2 $\binom{m}{j}$ changes sign at $j = \lfloor m \rfloor + 2$.

Proof. The Beta integral is the substitution $x = e^{-t}$ in $B(z + \alpha, \beta - \alpha)$. For (7.1), expand $(1 - q^{-u}) \sum_{k \geq 0} q^{k(d-u)}$; the coefficient of q^{-ku} is $q^{kd} - q^{(k-1)d} = q^{(k-1)d} (q^d - 1)$, and multiplicativity over the primes dividing n gives the Jordan totient. The last clause is Propositions 6.1 and 6.2. $\square$

Remark 7.6 (Reading). This is what the coincidence of intervals was hiding. The transfer's positivity was never inherited from a lattice: there is no proof of complete monotonicity that goes through one, and by Remark 3.4 a ratio in which the lattice cancels could not carry one. It comes from a Beta density and an Euler product, both of which take the rank as an exponent and therefore continue to every real value without noticing that a lattice has stopped existing. The point count takes the rank as a binomial index and notices immediately. The two positivity statements sit on the same family because the family has one parameter, and they part company because that parameter occupies a different position in each.

Corollary 7.7 (A factorization exhibiting the failure does not exist). *No factor of the decomposition $K_\omega = R_\omega K_\omega^\Gamma K_\omega^\zeta$ loses positivity at any real $d > 0$: by Proposition 7.1 each is positive throughout, and the only factor that changes character does so at $m = 2$, an integer, for a reason internal to $\Lambda(u) = \Lambda(1 - u)$. In particular there is no factorization of the transfer exhibiting which factor loses positivity at fractional d , because none does.*

8 Outlook

8.1 What is settled

Problem 1.1 is answered: $d = 2\omega$ is a parameter. The dictionary names one object, a rank- $(d+1)$ lattice (Section 3); the object exists at every integer and nothing strictly in between (Section 6); the transfer continues past the family's range with its decomposition and its positivity intact (Proposition 4.3) while its criterion does not (Proposition 5.1); and the transfer's positivity continues for reasons that have nothing to do with a lattice (Proposition 7.5), so the lattice's failure leaves no trace in it. There is no remaining sense in which d is a dimension away from the integers, and at fractional d there is a scattering matrix without a space.

Three questions that earlier stages of the investigation posed are closed negatively, and the closures are worth as much as the openings.

- *Does the criterion continue to the half-integer points beyond the range, where the object is a genuine lattice?* It continues, and the continuation is vacuous (Proposition 5.1). Testing contraction at $\omega = 1, \frac{3}{2}, 2$ cannot fail, so it calibrates the instrument and learns nothing about ζ .
- *Is the exchange of complete monotonicity for innerness at $m = 2$ the same fact as the point count's failure?* No (Corollary 7.3); the shared interval is the first gap (Remark 7.4).
- *Is there a factorization of the transfer that exhibits which factor loses positivity at fractional d ?* No (Corollary 7.7), because no factor loses positivity anywhere.

8.2 What is open

The complex-hyperbolic line. In the true half-distance the archimedean measure is the radial measure of complex hyperbolic space of complex dimension ω and real dimension 2ω , with Jacobi parameters $(\alpha, \beta) = (\omega - 1, 0)$ and $\rho = \omega > 0$ [6] — a Heckman–Opdam parameter, where a transform theory exists at arbitrary multiplicity [18]. Proposition 7.5 now explains why that half of the transfer continues at all: its positivity is a Beta density with ω in the exponent. The question is whether the Heckman–Opdam theory supplies anything beyond a name for that — an inversion, a Plancherel measure, a completeness statement that the Beta density does not already give. The honest prior is that it supplies a name.

Two conjectures about sums of squares. That $r_m(n) \geq 0$ for every n and every real $m \geq 4$; and that the negative coefficients on $m \in (3, 4)$ arise from competition between terms of both signs rather than from forcing (Remark 6.7). Neither is a question about ζ , and neither is used here.

A convention to retire. Remark 2.1: three research notes of this investigation write ξ for what this manuscript and its parent call Λ , and the two differ by exactly R_ω .

8.3 Status of every statement

- **Written proof, unconditional:** Propositions 3.1, 3.2, 4.2, 4.3, 5.1, 6.2, 6.3, 6.4, 6.5, 7.1, 7.5; and Corollaries 5.3, 6.6, 7.3, 7.7. None is conditional on the Riemann hypothesis and none mentions a zero except to bound its real part by the Euler product.

- **Written proof citing a classical theorem:** Proposition 6.4, whose last clause is Lagrange’s four-square theorem [13], used and not reproved; and Remark 5.2, which uses the classical zero-free region [17] at the single point $\omega = \frac{1}{2}$.
- **Inherited and used as stated:** Propositions 2.2, 2.3, 2.4, 2.5 and 4.1, all from [2]; and Proposition 6.1 from [7].
- **Registered computation:** the six programmes of Appendix A, 1522 finite test cases, each with a preserved record and each verified deterministic across repeated runs. Counts refer to test cases, not to independent theorems.
- **Calibration, not evidence:** every number in Table 4, for the reason in Remark 5.5.
- **Reading:** the group-theoretic sentence of Corollary 3.3; and Remark 7.6.
- **Conjecture, used nowhere:** the two statements of Remark 6.7.
- **Not claimed:** that any object exists at fractional d ; that the transfer is a contraction at any $\omega < \frac{1}{2}$; anything about the location of the zeros; any positivity of the Weil form.

A Programmes and records

Six programmes are *registered* in the sense of this program’s conventions: standard library only, JSON to standard output, a preserved record under `numerics/records/`, and registration in `validation/drafts.py`, so that `python3 validation/drafts.py check --replay` re-runs each and requires byte-identical output. There is no exploratory programme in this investigation. Nothing here is a positivity certificate, and counts refer to finite test cases rather than to independent theorems. Every value of ζ computed anywhere below is at a *real* argument, by a convergent series, a convergent theta integral or Euler–Maclaurin; no value off the real axis is computed and no zero is located.

check_dimension_dictionary.py (72 cases, four groups, under a second). The dictionary of [5]: the comb weights against the Jordan totient, $\tilde{c}_n = J_d(n)/n^{(d+1)/2}$, for $n < 120$ at seven shifts; $J_d(n)$ against brute-force counts of the primitive vectors in $(\mathbb{Z}/n\mathbb{Z})^d$ for $d \in \{1, 2, 3\}$, and $\tilde{c}_n = \varphi(n)/n$ at $d = 1$ in exact rationals; J_d as the Dirichlet coefficient of $\zeta(u - d)/\zeta(u)$ by Möbius convolution at six d , fractional ones included; and the reparametrization (2.5) including the Gamma factors and the Blaschke pole at $\frac{d+1}{2}$. *No value of ζ is computed in it.*

check_which_dimension.py (56 cases, five groups, under a second). Which coordinate carries the dimension [6]: $\tau = \log|t + i|$ against the hyperbolic distance $2 \operatorname{arcsinh}(t/2)$, confirming that τ is a Busemann function and $\tau \sim r^2/2$ a square; the same measure carried between coordinates, with normalized limits differing by $2^{1-\omega}$; the transfer integral rebuilt factor by factor in both readings, the Euclidean one reproducing the Gamma ratio to 3×10^{-11} ; the CH^ω radial measure in the true half-distance; and the Jacobi parameters in both coordinates against the group cases. *No value of ζ is computed in it.*

check_residue_volume.py (27 cases, four groups, under a second). The Blaschke residue as a volume term [6]: $\operatorname{Res}_{p=b} \bar{K}_\omega = |S^d|/2\zeta(d+1)$ in two forms at six d , agreeing to 10^{-12} ; the endpoint $6/\pi = 2/\operatorname{vol}$ and the classical Eisenstein residue; and $1/\zeta(m)$ as the primitive density in $\mathbb{Z}^m$ by brute-force counts of primitive points in balls, which is what fixes the rank at $d+1$. ζ is evaluated only at real $\sigma > 1$.

check_epstein_scattering.py (327 cases, eight groups, under a second). Section 3 and Section 6.1 [7]: Poisson summation in dimension m to 10^{-15} ; the completed Epstein integral against *direct lattice summation* with the exact continuum tail at $m = 2, 3, 4$ to 10^{-8} , which is the non-tautological test since that integral is symmetric in $s \leftrightarrow \frac{m}{2} - s$ by construction; (3.1) and the universality of the reflection ratio at $m = 2, \dots, 6$; the three expressions of (3.2) agreeing to 10^{-11} at $d = 1, \dots, 5$ with the reflection verified in exact rationals; the residue at $s = \frac{m}{2}$ by Richardson extrapolation; (6.2) against brute-force lattice enumeration in exact rationals with the degree bound by vanishing finite differences; the sign of $r_{d+1}(3)$ on a rational grid; the continuation of the cusp data; and the rank-two wall in exact rationals. ζ is evaluated at real arguments only, including negative ones, by a convergent theta integral.

check_two_blaschke.py (266 cases, six groups, about two seconds). Section 4 and Section 5.3 [8]. It is an extended copy of the parent’s registered assembly [2, App. A] with exactly one line changed — the removable 0/0 of Remark 5.4 — and a new driver; the change is marked in place. (A) the patched Gauss–Jacobi rule against exact moments at $\omega = \frac{1}{2}, 1, \frac{3}{2}, 2$, including $\omega = 1$ where the parent programme raises. (B) R_ω in three forms, unimodular on the axis at every ω , inner on a right half-plane grid exactly when $\omega \geq \frac{1}{2}$ and with a right half-plane pole at $p = a$ otherwise, and $K_\omega = R_\omega \widetilde{K}_\omega$ end to end. (C) the regrouped identity (4.3) against the parent’s grouping, the two parameter gaps, and the kernel sampled for positivity, at $\omega = 1$ against its closed form $2\pi e^{-\tau/2}$. (D) the assembly’s Laplace certificate at $\omega = 1, \frac{3}{2}, 2$ to 3×10^{-13} , with $\omega = 0.1, \frac{1}{2}$ as controls. (E) Table 4, with the quadrature drift from 32 to 48 nodes per atom. (F) the coincidence of the two intervals on a grid in d . ζ is evaluated only at real $u \geq 1.2$, in group (B), by Euler–Maclaurin.

check_the_trade.py (774 cases, seven groups, under a second). Sections 6 and 7 [9]. (A) $R_j(n)$ by dynamic programming against brute-force enumeration, $s(n)$, and the values $N_0 = 2, N_1 = 3, N_2 = 7$, with no N_k for $k \geq 3$ since $\max_{n \leq 240} s(n) = 4$. (B) the sign law (6.3) on six gaps, exact rationals. (C) the free region, the forced negative $r_m(N_k) < 0$ across the first three gaps, and the closed forms for $r_m(n)$, $n = 2, \dots, 7$, together with the representation data the hand bounds of Proposition 6.5 use. (D) no negative coefficient at integer $m = 1, \dots, 5$. (E) Proposition 7.1: a vanishing only at $m = 2$, $k_\omega^\Gamma > 0$, $\tilde{c}_n > 0$ and $J_d(n) > 0$ at ten values of m crossing every integer up to 6, and the $m = 3$ regularity threshold shown not to be a sign change. (F) the local Euler-factor coefficients (7.1) at five primes and seven real d , multiplying to the Jordan totient. (G) Table 5. *No value of ζ is computed anywhere in it.*

Two numerical lessons carried forward. The defect $\lambda_{\min}(D_N)$ is an absolute quantity of size 10^{-8} or smaller read off matrices of norm one, so only a few of its digits survive in double precision, and at $L = \log 5, \log 7$ it comes out of either sign at the 10^{-14} level: there the assertion made is $|||V_N|| - 1| < 10^{-12}$ and the floor is stated, rather than $\|V_N\| < 1$. And the single-atom kernel is $\tau^{\omega-1}$ times a function holomorphic on $|\tau| < \pi$, so the spike belongs in the Gauss–Jacobi *weight* and not in an evaluated quantity; that is what lets the whole assembly run in double precision [2, Sec. 5.1].

B Provenance, and the use of language models

This manuscript was drafted for Edward Baker by a language model of the Claude family (Anthropic), at his direction and in dialogue with him: working draft 0.1, on 18 September 2026, by Claude Opus 5 (model `claude-opus-5`). The investigation it reports was opened by Edward Baker on 18 September

2026 from the closed Loewner investigation [2], and the question it pursues — Problem 1.1, whether the dimension the transfer’s factors carry is a dimension or a parameter — is his.

The mathematical content consolidates five research notes of this investigation [5, 6, 7, 8, 9], each written by a model named at its head, together with the six numerical programmes and preserved records described in Appendix A, also written by them. Nothing is inherited without being restated: the propositions taken from the parent manuscript are listed in Section 2.3 and are used as stated, and every other proposition here is proved in place.

The proofs are short and are meant to be checked line by line by a human reader. The registered computations are replayed by `validation/drafts.py` and require byte-identical output; the readings, the calibration and the two conjectures are labelled as such in Section 8.3. Nothing in this document should be cited as established beyond what its own proofs establish.

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- [2] Parent manuscript of this investigation, *The Markov part of the shifted Weil transfer*, working draft 0.3, 18 September 2026, [papers/susy-positivity/investigations/loewner](#). Results are cited by their numbers in that draft.
- [3] Companion manuscript of this program, *Deformation flows for the shifted Weil family: the transfer as an all-pass filter*, working draft 0.8, 17 September 2026, [papers/susy-positivity/investigations/wilson-lines](#).
- [4] Research note of this program, *The shift is half a dimension: the archimedean factor is a horocycle integral, and the Loewner proposal closes*, 18 September 2026, [investigations/loewner/notes/THE_SHIFT_IS_HALF_A_DIMENSION_20260918.md](#).
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